

6. PROJECT DEVELOPMENT PHASE

Medical Diagnosis Center by Service Portal (ServiceNow)

The development phase records the practical implementation of the Medical Diagnosis Center application in the ServiceNow Developer Instance. The application is developed to manage patient information, diagnostic tests, appointments, and reports using ServiceNow.

The development flow includes application creation, table development, record management, business logic, and Service Portal implementation.

6.1 Phase 1 — Medical Diagnosis Center Application Creation

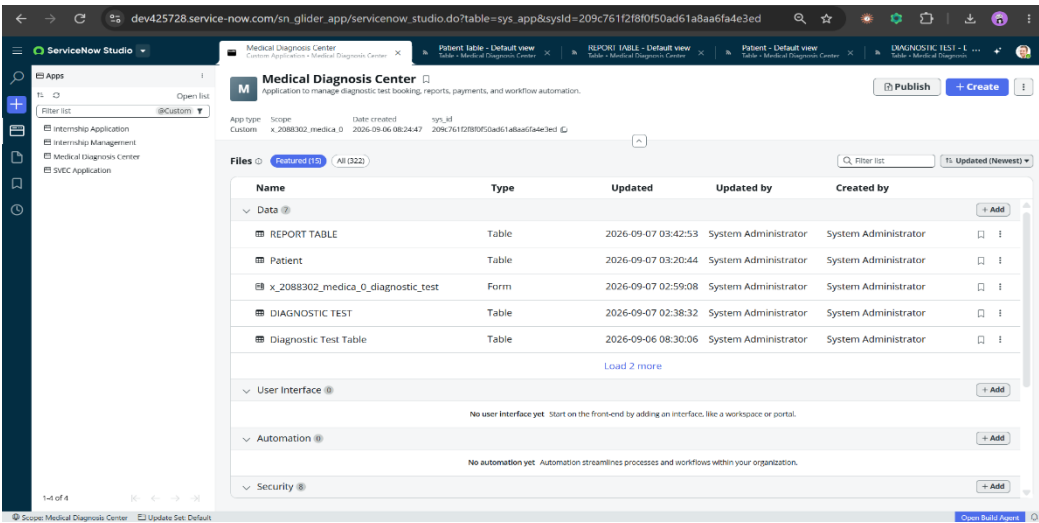
The Medical Diagnosis Center application is created in ServiceNow to provide a centralized environment for developing and managing the required application components.

Implementation

- Open ServiceNow Studio.
- Create the Medical Diagnosis Center application.
- Configure the application scope.
- Open the application in ServiceNow Studio.
- Verify the application files and data components.

Result: The Medical Diagnosis Center application is created successfully and is available for further development.

Figure 6: Medical Diagnosis Center Application



6.2 Phase 2 — Patient Table Creation

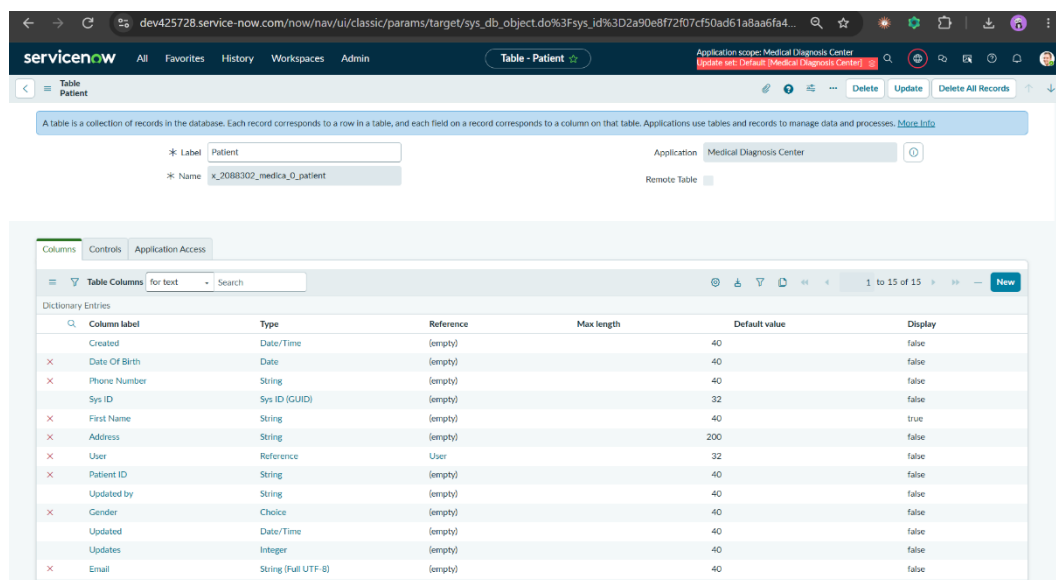
The Patient table is created to store and manage patient information required by the Medical Diagnosis Center.

Implementation

- Open the Medical Diagnosis Center application.
- Create the Patient table.
- Configure the required patient fields.
- Open the Patient table.
- Verify the table structure and records.

Result: The Patient table is created and available for storing patient information.

Figure 7: Patient Table



Column label	Type	Reference	Max length	Default value	Display
Created	Date/Time	(empty)	40		false
Date Of Birth	Date	(empty)	40		false
Phone Number	String	(empty)	40		false
Sys ID	Sys ID (GUID)	(empty)	32		false
First Name	String	(empty)	40		true
Address	String	(empty)	200		false
User	Reference	User	32		false
Patient ID	String	(empty)	40		false
Updated by	String	(empty)	40		false
Gender	Choice	(empty)	40		false
Updated	Date/Time	(empty)	40		false
Updates	Integer	(empty)	40		false
Email	String (Full UTF-8)	(empty)	40		false

6.3 Phase 3 — Patient Record Management

The Patient record is used to maintain individual patient information in the Medical Diagnosis Center application.

Implementation

- Open the Patient table.
- Open the required Patient record.
- Verify the configured patient fields.
- Enter or update the required patient information.
- Save the patient record.

Result: Patient information is stored and maintained successfully in the application.

Figure 8: Patient Record

The screenshot shows the ServiceNow 'Patient - New Record' form. The top navigation bar includes 'servicenow', 'All', 'Favorites', 'History', 'Admin', and a 'Patient - New Record' button. The application scope is 'Medical Diagnosis Center'. The form contains several input fields: Patient ID, First Name, Phone Number, Email, Last Name, Gender (a dropdown menu currently showing '-- None --'), Date Of Birth (with a calendar icon), User (with a search icon), and Address. A 'Submit' button is located at the bottom left of the form area.

6.4 Phase 4 — Diagnostic Test Table Creation

The Diagnostic Test table is created to maintain the diagnostic tests available in the Medical Diagnosis Center.

Implementation

- Open the Medical Diagnosis Center application.
- Create the Diagnostic Test table.
- Configure the required diagnostic test fields.
- Open the Diagnostic Test table.
- Verify the available diagnostic test records.

Result: The Diagnostic Test table is created and available for maintaining diagnostic test information.

Figure 9: Diagnostic Test Table

Table - DIAGNOSTIC TEST

Application scope: Medical Diagnosis Center
Update with Default (Medical Diagnosis Center)

* Label: **DIAGNOSTIC TEST**
* Name: **x_2088302_medica_0_diagnostic_test**

Application: Medical Diagnosis Center
Remote Table:

Columns | Controls | Application Access

Table Columns for text Search

Dictionary Entries

Q	Column label	Type	Reference	Max length	Default value	Display
	Created	Date/Time	(empty)	40		false
	Updates	Integer	(empty)	40		false
×	Price	Currency	(empty)	20		false
	Updated by	String	(empty)	40		false
×	Description	String	(empty)	200		false
×	Test Duration	Integer	(empty)	255		false
	Updated	Date/Time	(empty)	40		false
×	Test Code	String	(empty)	40		false
×	Test Name	String	(empty)	40		false
	Sys ID	Sys ID (GUID)	(empty)	32		false
×	Active	True/False	(empty)	255	True, False	false
✓	Created on	DateTime	timestamp	40		false

6.5 Phase 5 — Diagnostic Test Record Management

The Diagnostic Test records are used to maintain information about the tests provided by the Medical Diagnosis Center.

Implementation

- Open the Diagnostic Test table.
- Open the required diagnostic test record.
- Verify the configured test information.
- Enter or update the required details.
- Save the diagnostic test record.

Result: Diagnostic test information is successfully maintained in the application.

Figure 10: Diagnostic Test Record

The screenshot shows the 'servicenow' interface for creating a new 'DIAGNOSTIC TEST' record. The top navigation bar includes 'All', 'Favorites', 'History', and a 'DIAGNOSTIC TEST - Create Created' button. The form itself has the following fields:

- Test Name:** A text input field.
- Test Code:** A text input field.
- Category:** A dropdown menu currently showing '-- None --'.
- Active:** A checkbox.
- Test Duration:** A text input field.
- Price:** A field with a currency symbol '\$' and a value of '0.00'.
- Description:** A large text area.

A 'Submit' button is located at the bottom left of the form area.

6.6 Phase 6 — Report Table Creation

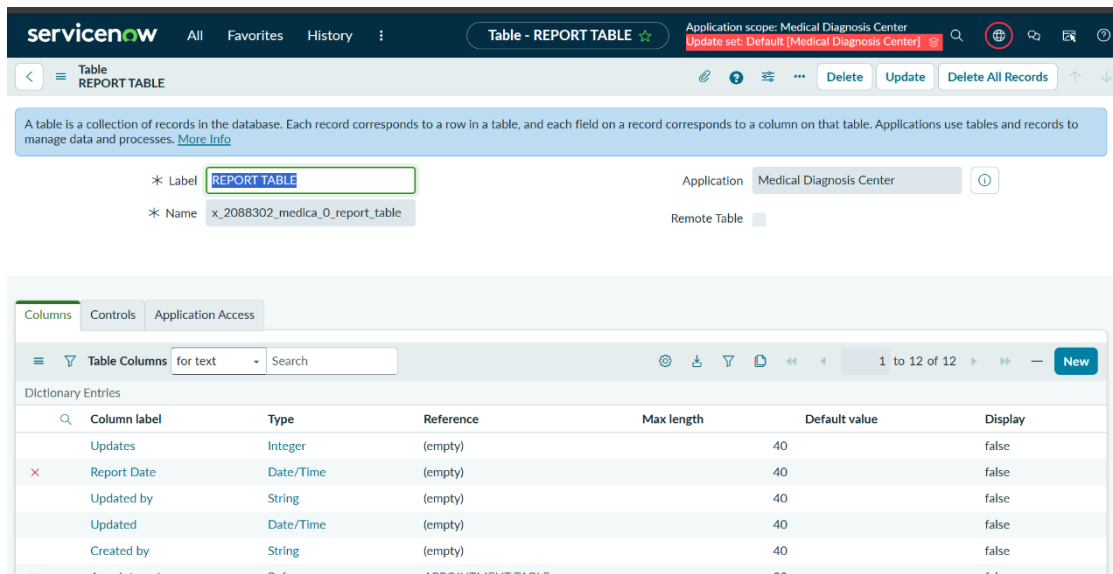
The Report table is created to maintain diagnostic reports generated for the Medical Diagnosis Center.

Implementation

- Open the Medical Diagnosis Center application.
- Create the Report table.
- Configure the required report fields.
- Open the Report table.
- Verify the report records.

Result: The Report table is created and available for managing diagnostic report information.

Figure 11: Report Table



6.7 Phase 7 — Application Data Validation

The application data components are checked to confirm that the required tables have been created under the Medical Diagnosis Center application.

Implementation

- Open the Medical Diagnosis Center application in ServiceNow Studio.
- Open the Data section.
- Verify the Patient table.
- Verify the Diagnostic Test table.
- Verify the Report table.
- Check the available application files.

Result: The required application tables are available under the Medical Diagnosis Center application.

Figure 12: Medical Diagnosis Center Application Data

ServiceNow Studio

Apps

Filter list

Open list

@Custom

Internship Application

Internship Management

Medical Diagnosis Center

SVEC Application

Patient Table - Default view

REPORT TABLE - Default view

Patient - Default view

Table • Medical Diagnosis Center

Medical Diagnosis Center

Preview

Save

Table fields

Add new field

Column label *	Column name *	Type *	Reference	Max length
Address	address	String		200
Created	sys_created_on	Date/Time		
Created by	sys_created_by	String		40
Date Of Birth	date_of_birth	Date		
Email	email	String (Full UTF-8)		40
First Name	first_name	String		40
Gender	gender	Choice 3 Choices		
Last Name	last_name	String		40

Add new field

Showing 1-14 of 14

Scope: Scope is controlled & indicated by the builder.

Open Build Agent

ServiceNow Studio

Apps

Filter list

Open list

@Custom

Internship Application

Internship Management

Medical Diagnosis Center

SVEC Application

REPORT TABLE - Default view

Patient - Default view

DIAGNOSTIC TEST - Default view

Table • Medical Diagnosis Center

Table • Medical Diagnosis Center

Table • Medical Diagnosis Center

Medical Diagnosis Center

Preview

Save

Table fields

Add new field

Column label *	Column name *	Type *	Reference	Max length	Default
Active	active	True/False			True,
Category	category	Choice 8 Choices			
Created	sys_created_on	Date/Time			
Created by	sys_created_by	String		40	
Description	description	String		200	
Price	price	Currency			
Test Code	test_code	String		40	
Test Duration	test_duration	Integer			

Add new field

Showing 1-12 of 12

Scope: Scope is controlled & indicated by the builder.

Open Build Agent

ServiceNow Studio

Apps

Filter list

Open list

@Custom

Internship Application

Internship Management

Medical Diagnosis Center

SVEC Application

REPORT TABLE - Default view

Patient Table - Default view

REPORT TABLE - Default view

Table • Medical Diagnosis Center

Table • Medical Diagnosis Center

Table • Medical Diagnosis Center

Medical Diagnosis Center

Preview

Save

Table fields

Add new field

Column label *	Column name *	Type *	Reference	Max length
Appointment	appointment	Reference	APPOINTMENT T...	
Created	sys_created_on	Date/Time		
Created by	sys_created_by	String		40
Findings	findings	HTML		
Patient	patient	Reference	Patient	
Report Date	report_date	Date/Time		
Report File	report_file	File Attachment		
Updated	sys_updated_on	Date/Time		

Add new field

Showing 1-11 of 11

Scope: Scope is controlled & indicated by the builder.

Open Build Agent

6.8 Phase 8 — Appointment Table and Record Development

The Appointment table is used to manage appointments between patients and the Medical Diagnosis Center.

Implementation

- Open the Appointment table.
- Configure the required appointment information.
- Select the required patient.
- Select the required diagnostic test.
- Enter the appointment details.
- Save the appointment record.

Result: The appointment record is created with the required patient and diagnostic test information.

Figure 13: Appointment Record

The screenshot shows the ServiceNow 'Appointment Table - Create New Record' form. The form is titled 'APPOINTMENT TABLE - Create New Record' and has a 'Submit' button at the bottom right. The form is divided into two columns. The left column contains the following fields: Appointment Number (004), Patient (Phani Teja Sri), Status (InProgress), Approval Status (Pending), Email (phanimotupalli2006@gmail.com), Approval Date (2026-09-19 22:38:57), Test, Scheduled Date (2026-09-21 22:39:12), and Slot Time (09:00 AM). The right column contains the following fields: Rejection Reason, Confirmation Sent (checked), Created By (System Administrator), Approved By (System Administrator), Notes, Payment Status (Pending), Total Amount (\$ 20.00), and Refund Amount (\$ 10.00). A 'Submit' button is located at the bottom left of the form.

6.9 Phase 9 — Appointment Cancellation and Refund

The appointment cancellation process is used to cancel an existing appointment and calculate the applicable refund according to the project requirements.

Implementation

- Open the required appointment.
- Select the cancellation option or update the appointment status.
- Save the cancelled appointment.

- Verify the refund calculation.
- Verify the updated appointment information.

Result: The appointment is cancelled and the required refund information is calculated.

Figure 14: Appointment Cancellation and Refund

[Insert your actual cancellation/refund screenshot here.]

6.10 Phase 10 — Business Rules and Automation

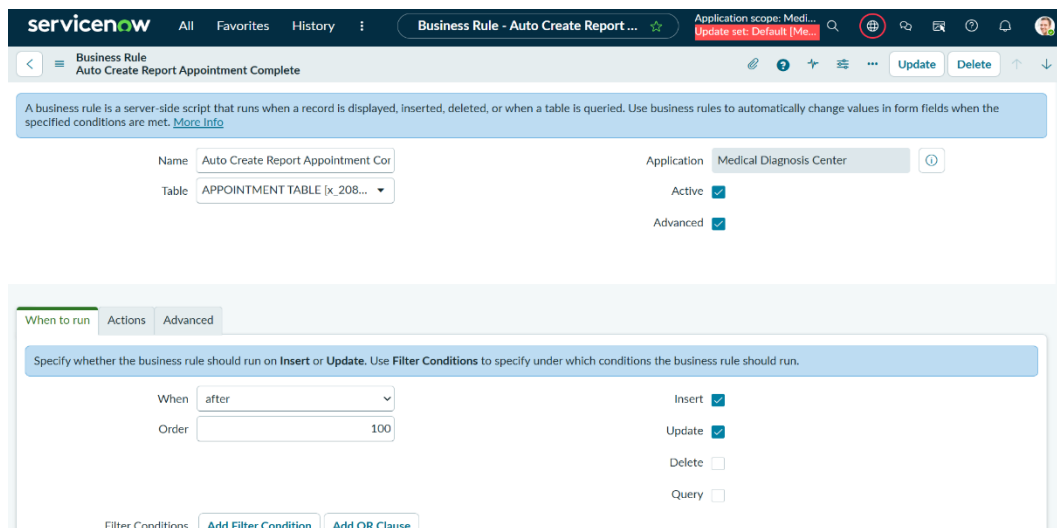
Business Rules and automation are configured to support the required Medical Diagnosis Center processes.

Implementation

- Open the Medical Diagnosis Center application.
- Open the required Business Rule or automation configuration.
- Configure the required conditions and actions.
- Save the configuration.
- Verify the automated behavior.

Result: The required business logic and automation are configured for the application.

Figure 15: Business Rules and Automation



6.11 Phase 11 — Service Portal Development

The Service Portal provides users with an interface to access the Medical Diagnosis Center services.

Implementation

- Open the Service Portal configuration.
- Configure the Medical Diagnosis Center portal.
- Configure the required portal pages.
- Add the required components or widgets.
- Open the portal to verify the user interface.

Result: The Medical Diagnosis Center Service Portal is configured and available for users.

Figure 16: Medical Diagnosis Center Service Portal

The screenshot shows the ServiceNow configuration interface for the Medical Diagnosis Center Service Portal. The top navigation bar includes 'servicenow', 'All', 'Favorites', 'History', and a search bar. The main header displays 'Service Portal - Medical Diagnostic ...' and 'Application scope: Medi...'. A blue notification bar at the top states: 'Please add the field 'Enable certificate based authentication' using Form Layout to enable the feature. [More Details](#)'. The configuration form includes fields for:

- * Title: Medical Diagnostic Center
- * URL suffix: mdc
- Homepage: test_list
- KB home page
- Login page
- Logo: Click to add...
- Logo /Alt Text: Medical Diagnostic Center
- Icon: Click to add...
- Application: Medical Diagnosis Center
- 404 page
- Catalog home page
- Catalog category home page
- Main menu
- Theme: Stock
- Inactive: ☐
- Quick start config:
- CSS variables: `-webkit-text-fill-color: #ffffff !important;` and `box-shadow: 0 2px 8px rgba(230, 57, 70, 0.5) !important;`

6.12 Phase 12 — Appointment Booking Through Service Portal

The appointment booking process allows the user to submit the required appointment information through the Service Portal.

Implementation

- Open the Medical Diagnosis Center Service Portal.
- Open the appointment booking page.
- Select the required patient.
- Select the required diagnostic test.
- Enter the appointment details.
- Submit the appointment.

Result: The appointment is created through the Service Portal.

Figure 17: Appointment Booking Through Service Portal

6.13 Phase 13 — Appointment Completion and Report Generation

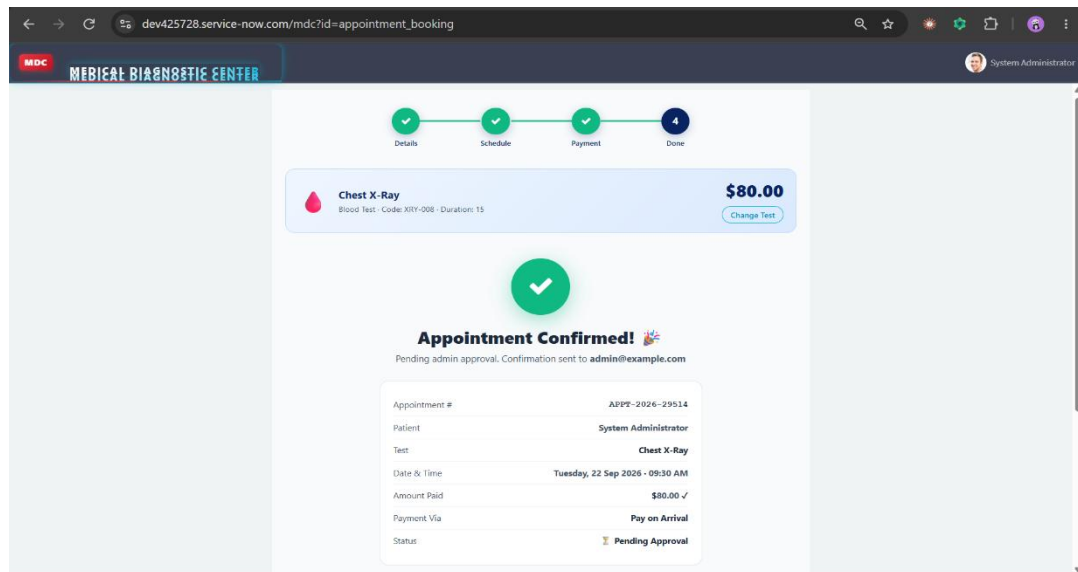
The report generation process is used to create a diagnostic report after the appointment is completed.

Implementation

- Open the required appointment.
- Verify the appointment information.
- Complete the appointment.
- Allow the configured automation to process the completed appointment.
- Open the generated report.
- Verify the report information.

Result: The diagnostic report is generated for the completed appointment.

Figure 18: Appointment Completion and Report Generation



6.14 Phase 14 — Final Validation

The completed application is checked to confirm that the main records and relationships are available and working as required.

The following functions are verified:

- Patient information is available.
- Diagnostic test information is available.
- Appointment records can be created and managed.
- Appointment cancellation is supported.
- Refund calculation is performed as required.
- Completed appointments generate reports.
- Report records can be managed.
- Service Portal functionality is available.

Result: The Medical Diagnosis Center application is validated against the implemented requirements.

6.15 Testing and Deployment Validation

The Medical Diagnosis Center application is tested to confirm that the implemented functionality works correctly.

Implementation

- Verify Patient table and records.
- Verify Diagnostic Test table and records.
- Verify Appointment functionality.
- Verify cancellation and refund functionality.
- Verify report generation.
- Verify report management.
- Verify Service Portal functionality.

Result: The implemented Medical Diagnosis Center functionality is tested and validated in the ServiceNow Developer Instance.

6.16 Conclusion

The Medical Diagnosis Center application is developed in ServiceNow to provide a structured solution for managing patient information, diagnostic tests, appointments, cancellations, refunds, and diagnostic reports.

The developed application uses ServiceNow tables, records, business logic, automation, and Service Portal components to support the required Medical Diagnosis Center processes.